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Conversations with Copilot ..... **Error! Bookmark not defined.**

Introduction

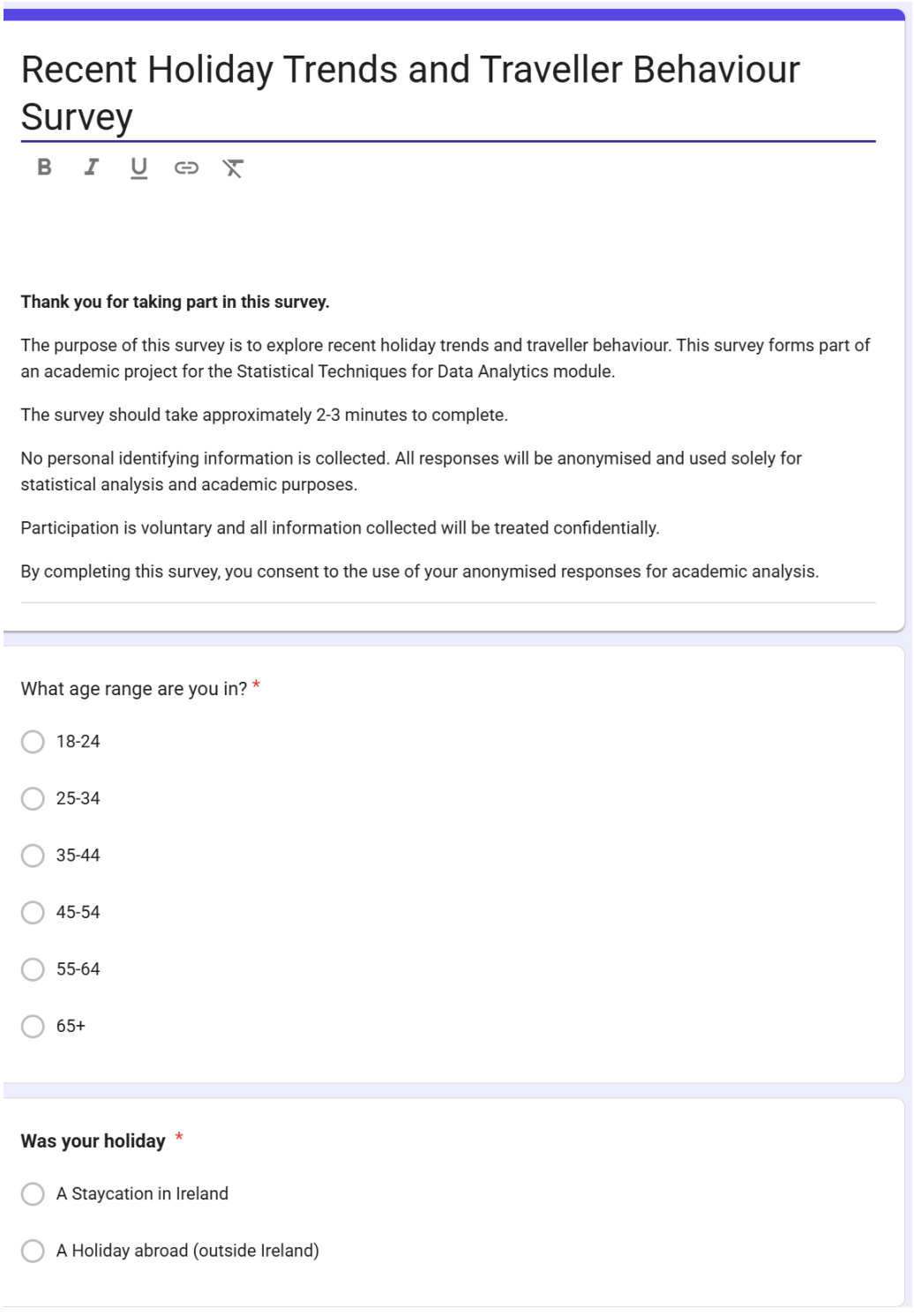
The purpose of this study was to investigate holiday trends and traveller behaviour. With today's cost of living issues, I wanted to gauge if people are still willing to take summer holidays and get an understanding of the overall costs involved.

As part of the survey, I wanted to find out if people are still going overseas or preferring to stay in Ireland. Other areas of the survey were to look at the age range of respondents to get an understanding of how they book holidays and how much on average the holiday costs (overall and accommodation costs).

The survey was designed with Google Forms and called:

Recent Holiday Trends and Traveller Behaviour Survey

A link to the survey was issued to friends, family and fellow students on my Phone contacts. Everyone was notified that this survey was for academic purposes only and all responses will be anonymised.



Recent Holiday Trends and Traveller Behaviour Survey

Thank you for taking part in this survey.

The purpose of this survey is to explore recent holiday trends and traveller behaviour. This survey forms part of an academic project for the Statistical Techniques for Data Analytics module.

The survey should take approximately 2-3 minutes to complete.

No personal identifying information is collected. All responses will be anonymised and used solely for statistical analysis and academic purposes.

Participation is voluntary and all information collected will be treated confidentially.

By completing this survey, you consent to the use of your anonymised responses for academic analysis.

What age range are you in? *

- 18-24
- 25-34
- 35-44
- 45-54
- 55-64
- 65+

Was your holiday *

- A Staycation in Ireland
- A Holiday abroad (outside Ireland)

Figure 1 shows a snapshot of the Google Form

Data Preparation

The questions were designed in a way to give us data on:

- Categorical Variables (Age Range, Holiday Location)
- Numerical Variables (Accommodation Costs)
- Likert Scale (Satisfaction)
- Boolean Variables (Booking Method)

Data Preparation Actions

Once completed, I received 71 responses on Google Forms from my survey. This survey was downloaded as a CSV file and named:

D**ata_CA1.csv**

The next stage was to clean the data, such as:

- Trim any leading spaces from the form
- Check for null values
- Check shape
- Check the information on loaded data
- Rename columns to make it more readable

No missing values were discovered on the dataset and was now ready for analysis.

Descriptive Analysis

Descriptive statistics were carried out on the survey dataset to identify patterns from respondents.

Descriptive Statistics

```
df_Holidays.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Group_Size	71.0	3.253521	1.754328	1.0	2.0	3.0	4.0	10.0
Accommodation_Cost	71.0	198.295775	478.726686	1.0	80.0	115.0	150.0	4000.0
Holiday_Length	71.0	7.267606	5.022685	1.0	4.0	6.0	7.5	28.0
Total_Cost	71.0	2285.915493	1611.923941	200.0	850.0	2000.0	3000.0	6000.0
Satisfaction	71.0	4.478873	0.673141	3.0	4.0	5.0	5.0	5.0
Holidays_Last_12_Months	71.0	3.000000	3.171300	1.0	2.0	2.0	3.0	22.0

Figure 2 Shows Descriptive Statistics

The average Accommodation cost (per night) was €198.29. However, the median analysis shows the accommodation cost of €115.00. This would suggest that the distribution is positively skewed due to a few unusually high accommodation costs. This is further supported with a max value of €4,000.

Skewness analysis was presented to understand the shape of the distributions. As expected, Accommodation Cost displayed a high positive skewness of 7.43, confirming a few extreme outliers.

The mean Holiday length showed as 7.26 days, whilst it showed a positive skewness of 2.55, again resulting an outlier, with a max value of 28 days, influencing the overall distribution.

For a categorical Analysis, an age bar chart that was created shows that respondents were reasonably evenly distributed across the younger and middle-age categories. The 25–34 and 35–44 age groups represented the largest proportion of respondents, while the 65+ category contained the fewest responses.

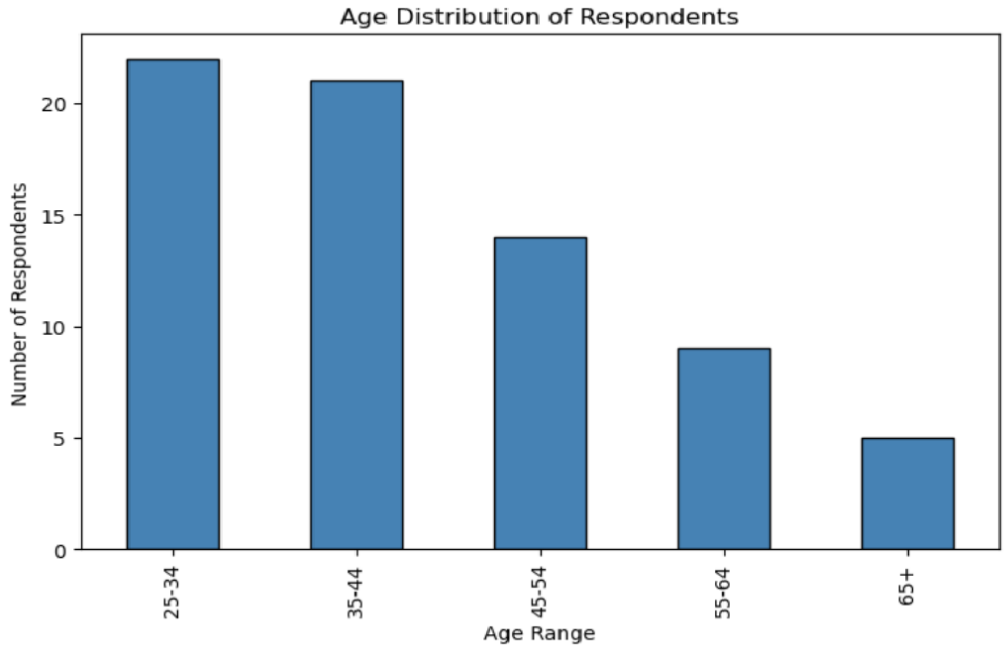


Figure 3 Shows the Age Distribution Bar Chart

Online bookings were the most popular method, with approximately 78% of all bookings, as opposed to package holidays.

Beach holidays abroad were the most popular holiday type. This is not surprising as the survey took place immediately after the summer holiday season.

```
df_Holidays['Holiday_Purpose'].value_counts()
```

```
Holiday_Purpose
Beach Holiday abroad                              37
City Break                                        11
To visit family / Friends                         11
Other                                              7
To attend an event (concert, Football match etc)   5
Name: count, dtype: int64
```

Figure 4 Shows the breakdown of Holiday purpose

Overall, the descriptive statistics provided a better understanding of holiday spending patterns, satisfaction ratings and holiday types. However, we can see a few outliers in the dataset that warranted further investigation.

Analysis & Insights

Booking Methods by Age Range

See below: pivot table was created to give an analysis of Booking methods by age range.

Run a pivot table to show Booking Methods broken down by Age Range

[59]:
```
# create a pivot table
pivot_bookings = pd.pivot_table (
    df_Holidays,
    values="Group_Size",
    index="Booking_Method",
    columns="Age_Range",
    aggfunc="count",
    margins=True,
    margins_name="Total"
)
pivot_bookings
```

[59]:

Age_Range	25-34	35-44	45-54	55-64	65+	Total
Booking_Method						
A package holiday	1	4	5	4	1	15
Independently Booked online	21	17	9	5	4	56
Total	22	21	14	9	5	71

Figure 5 Shows the pivot table of booking methods by age range

By breaking down Booking Methods by age range, we can see that online bookings remained the preferred choice of booking across all age categories, although package holidays were more common among the respondents aged 45-54.

Relationship to Holiday Duration, Total Expenditure & overall Satisfaction

In this analysis, a scatterplot was used to visualise this multivariate analysis

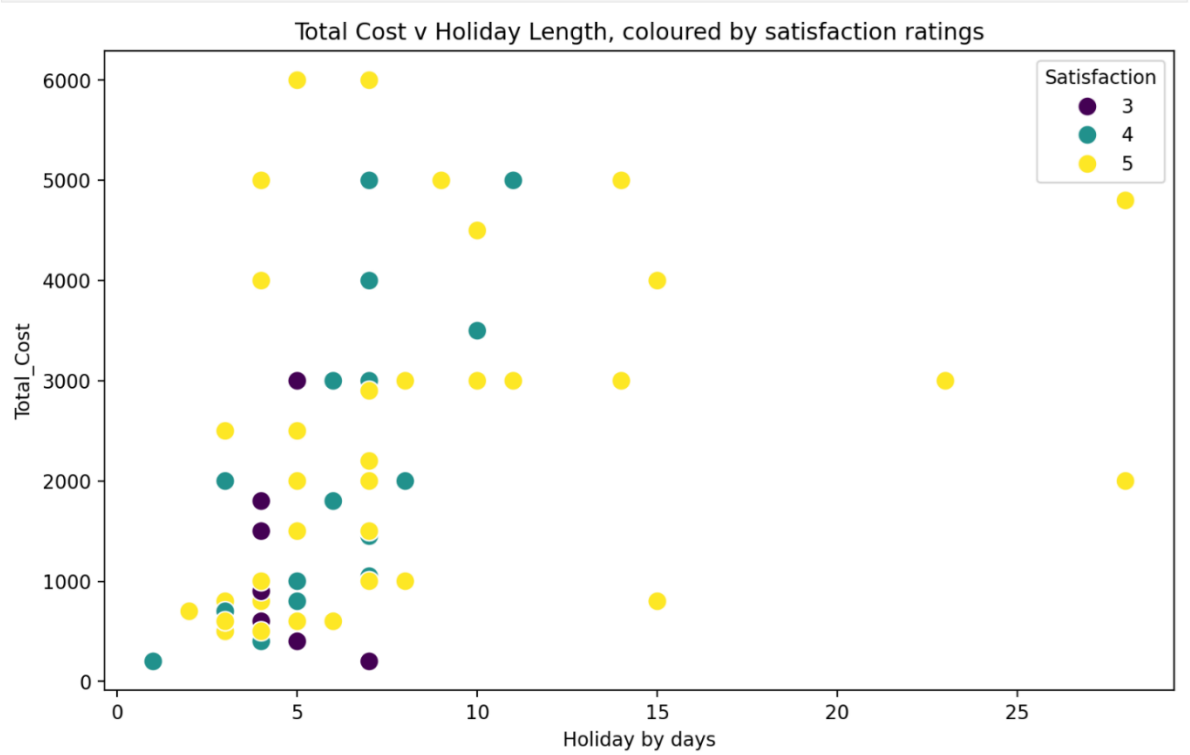


Figure 6 Scatterplot of Total Cost, Holiday Duration and Satisfaction

Scatterplot Analysis

The scatterplot shows a positive relationship between duration of holiday and total cost. However, a higher satisfaction rating can be observed across both low-cost and high-cost holidays.

Probability

A probability measurement was gathered to search the booking methods of the 25-34 age range. The prior probability shows:

P(A) Probability respondent is aged 25-34 = 30.99%

P(B) Probability respondent booked online independently = 78.87%

The conditional probability result of 37.5% was received when considering that a respondent was 25-34 when considering only the booking method.

Bayes Theorem also produced a 37.5% result. This was higher than the overall proportion of respondents aged 25-34, suggesting that this age group was more likely to book holidays online.

Correlation and Outlier Analysis

A streamlined data frame was created along with a new derived variable called "Cost_Per_Night". This was created by dividing Total_Cost by Holiday_Length and then dividing by Group_Size.

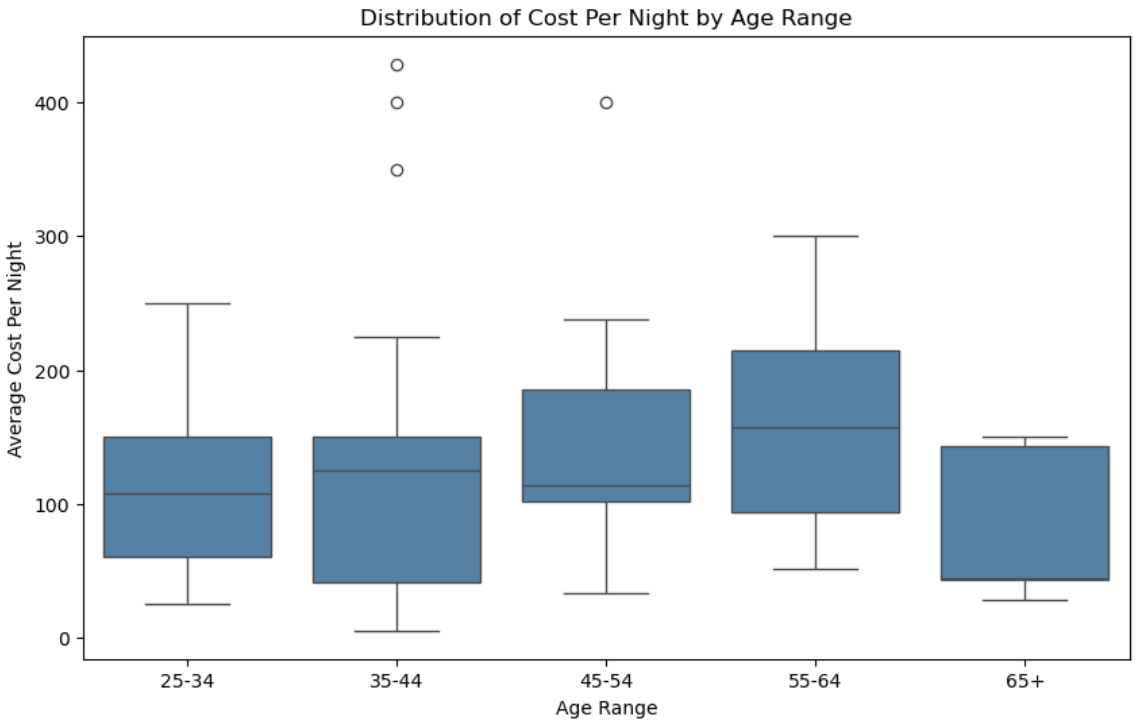


Figure 7 Boxplot showing Outlier Analysis

The presence of outliers helps explain why some age groups have noticeably higher mean values in the previous bar chart

The histogram of Cost_Per_Night demonstrates a positively skewed distribution. Most respondents recorded a cost per night below €300, whilst a smaller number of respondents reported a significantly higher spending. The presence of these higher values creates a longer tail on the right-hand side of the distribution, indicating potential outliers.

Correlation Analysis

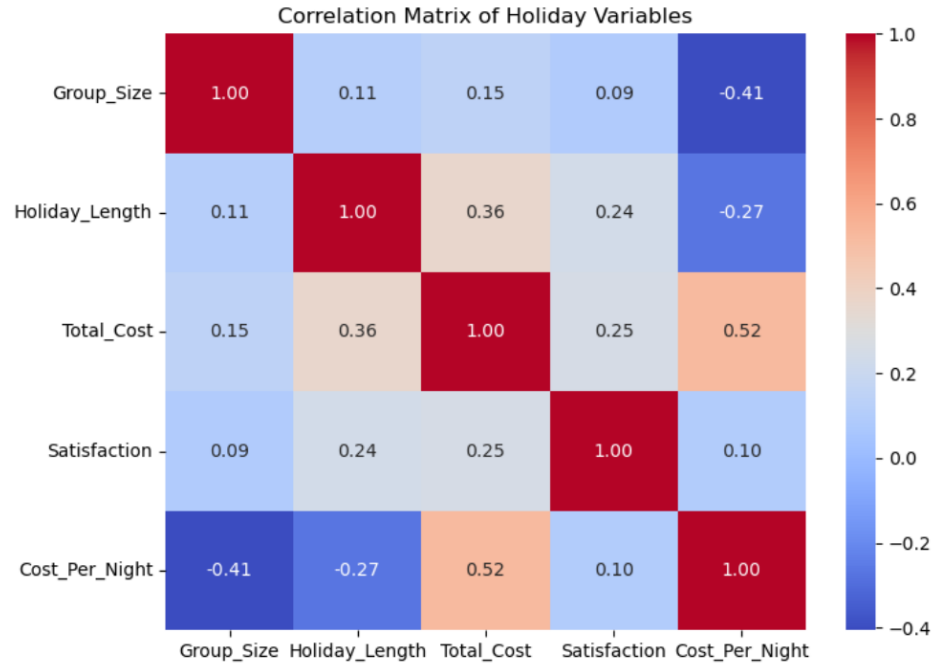


Figure 8 Correlation Matrix

The strongest positive correlation was between Total_Cost and Cost_Per_Night (r=0.52). This indicates that respondents who spent more overall also tended to spend more on a per-night basis, which is to be expected.

Outlier Analysis using Accommodation Costs

The distribution clearly shows the presence of several extreme accommodation costs.

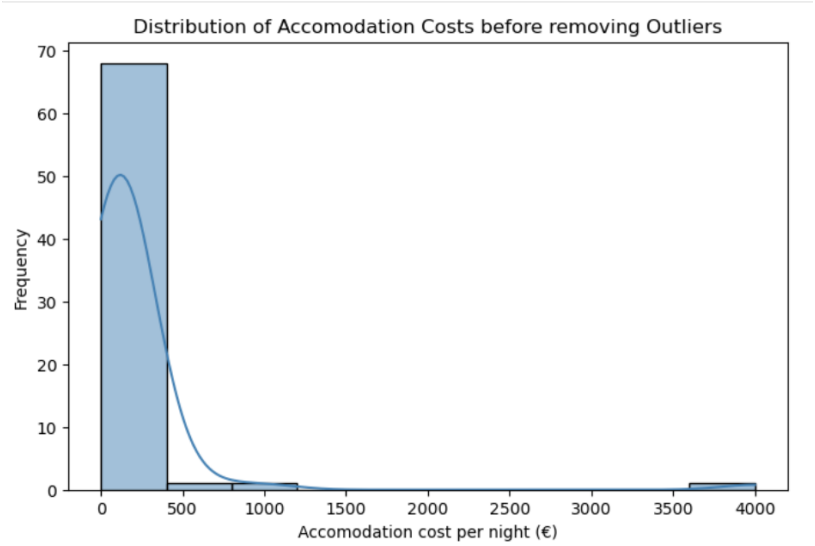


Figure 9 Distribution of Accommodation costs before outliers were removed

The mean accommodation costs of €111.39 are recorded after removal of outliers. The original dataset has a mean of €198.28 before removal of outliers. This further indicates that a small number of observations were increasing the average cost.

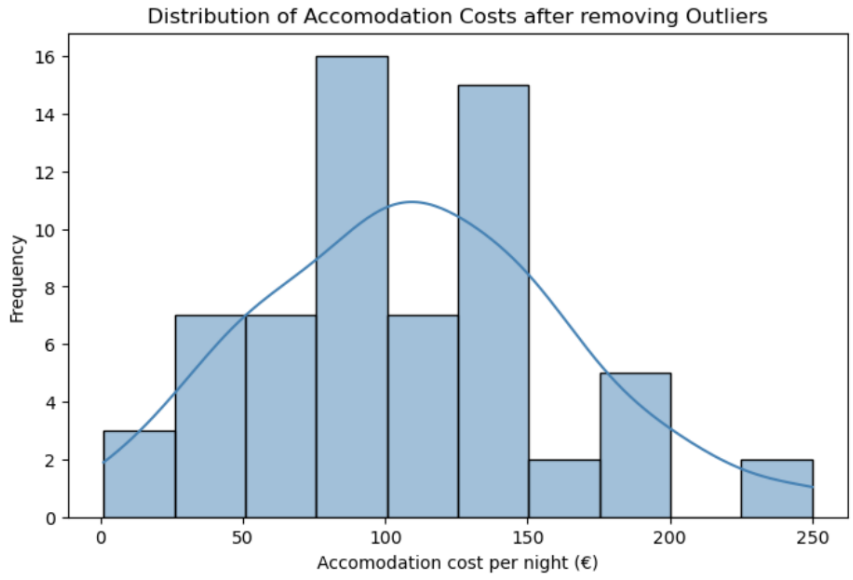


Figure 10 Distribution of Accommodation Costs after outliers were removed

Using the IQR Method to identify outliers with the Accommodation Cost variable

- Q1 was 80
- Q3 was 150
- IQR was 70

By applying the standard 1.5 x IQR the upper bound was €255 and the lower bound was €-25 were calculated. Therefore, only Rows exceeding the upper bounds were classified as outliers and removed. This resulted in seven observations classified as outliers and removed, with a new dataset of 64 remaining.

A normality test was carried out on the accommodation cost, before and after outliers were removed. Once removed, a p-value of 0.159 was returned, showing that the distribution appeared to be more normally distributed, whereas before removing outliers, it did not appear to be normally distributed,

Conclusion and Discussion

The analysis provided valuable insights into the Holiday Trends survey, showing us that online bookings were the preferred method across all age groups, while longer holidays resulted in higher expenditure. Satisfaction levels were high across the dataset, regardless of expenditure.

The impact of outliers on the Accommodation Cost variable was a main finding. Extreme values resulted in a positively skewed distribution. Once outliers were removed, using the IQR method, the distribution became more representative of typical accommodation costs.

Using Python and its libraries, I was able to take a raw survey dataset and transform it into meaningful insights, through statistical analysis and data visualisation techniques, enabling trends and unusual observations to be identified.

Word count = 1120

References

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Appendix

Appendix A – Reflection on the use of Artificial Intelligence

Artificial Intelligence (Microsoft 365 Copilot) was used throughout this project as a support tool to assist with understanding Python programming, report planning and guidance in developing Google Forms. AI was primarily used to help explain correlation analysis and outlier detection, although I have previous experience with these, whilst studying my Level 7 in Data Technology.

AI was also used to provide guidance when debugging code errors in Python, such as plotting visualisations, correlation matrices and pivot tables. In addition, AI provided suggestions on how to structure the reporting and improve the organisation of Jupyter Notebook.

A major benefit of AI was the ability to explain concepts and provide examples that could be adapted to my survey dataset. AI also helped identify areas where additional analysis was required, such as correlation analysis and outlier investigation.

However, all survey data was collected and analysed independently. The final decisions regarding the choice of statistical techniques, interpretation of results and conclusions remained my own responsibility.

Overall, the use of AI enhanced my learning experience by acting as a study aid and technical assistant, improving both my understanding of statistical techniques and my confidence in applying Python for data analysis.

AI was also used to review draft interpretations and suggest alternative wording for statistical findings. These suggestions were reviewed critically before being incorporated into the final report